

Total No. of Questions : 8]

SEAT No. :

PD-4076

[Total No. of Pages : 2

[6402]-36

S.E. (Computer Engineering/Computer Science & Design/AI & DS)

FUNDAMENTALS OF DATA STRUCTURE

(2019 Pattern) (Semester - III) (210242)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Draw neat & labelled diagrams if necessary.

Q1) a) Discuss an algorithm for sentinel search. Compare linear search & sentinel search. Comment on time & space complexity of both. [7]

b) Explain quick sort & sort the given list using quick sort. [7]

54, 26, 93, 17, 77, 31, 44, 50, 20. Analyze time complexity of quick sort.

c) Differentiate between internal and external sort. [3]

OR

Q2) a) Sort the following numbers using insertion sort : [7]

4, 3, 2, 10, 12, 1, 5, 6, 76, 36

Comment on efficiency, stability, in-place characteristics of insertion sort.

b) Enlist non-comparison based sorting techniques. Explain bucket sort with suitable example. Comment on time and space complexity of it. [7]

c) Apply sentinel search to find key = 39. [3]

34, 52, 78, 33, 67, 12, 45, 23

Q3) a) Represent following polynomial using generalized linked list $((x^{12} + 2x^9)y^4 + 4x^9y^2)z^3 + ((x^5 + 6x^3)y^5 + 3y)z$ [8]

b) Write an algorithm to perform following operations on singly linked list. [9]

i) Reverse

ii) Merge

P.T.O.

OR

- Q4) a)** What is linked list? Enlist different types of linked list. Write pseudo C/C++ code to insert a node in a doubly linked list (3 cases). [8]
b) Explain polynomial addition using SLL with suitable example and write pseudo C/C++ code for polynomial addition using singly linked list. [9]

- Q5) a)** How to convert infix expression into postfix expression using stack & convert following expression into postfix form. (Show all intermediate steps) $A * (B - C) / E \wedge F + G$ [9]
b) Write an algorithm for converting prefix expression into postfix expression. Convert following expression into postfix. $(A + B) - (C/D - E) + (F + G/K)$ [9]
c) Write an algorithm for converting prefix expression into postfix expression. Convert following expression into postfix. $[4 + 5 = 9]$
 $(A + B) - ((C/D - E) + (F + G/K))$

OR

- Q6) a)** What is stack? Write an ADT for stack and its implementation using array. [6 + 3 = 9]
What are different applications of stack?
b) What are polish notations? What is need of it?
Evaluate below prefix and postfix expression for $a = b = c = 2$ and $d = 1$
Prefix Expression = $- + a * bcd$
Postfix Expression = $abc * + d -$ [3 + 6 = 9]

- Q7) a)** What is queue? How they are represented in memory?
write a pseudocode to implement insert & delete operation in a linear queue using array. [3 + 6 = 9]
b) What is linked queue?
Write a function for inserting & deleting a node in a linked queue. [3 + 6 = 9]

OR

- Q8) a)** Explain the concept of linear queue and circular queue. Give the advantages of circular queue over linear queue.
Write C/C++ code to implement enqueue & dequeue operation on circular queue. [4 + 5 = 9]
b) What is Deque? Explain operations of Deque.
Write C/C++ code for insertion & deletion operations of it using array. [4 + 5 = 9]

